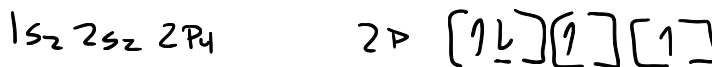


1. = Aktiv Chemistry Ch 3 # 77. Write the complete ground-state electron configuration (i.e. H: $1s^1$) for neon (Ne). (1 pt)



2. = Aktiv Chemistry Ch 3 # 97. How many unpaired electrons are found in a ground-state atom of oxygen (O)? (1 pt)



3. = Aktiv Chemistry Ch 3 # 105. Which element has the ground-state abbreviated electron configuration: $[Ar]4s^2 3d^{10} 4p^5$? (1 pt) Circle the answer below.

a. As b. Br c. Se d. Te e. I

4. = Aktiv Chemistry Ch 3 # 105. Rank the following atoms in order of decreasing size (i.e., largest to smallest): Rb, F, Mg, B, N. (1 pt) Circle the answer below.

A) Rb > Mg > B > N > F

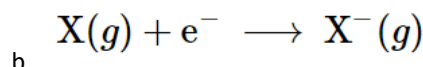
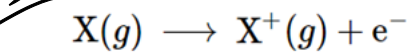
B) N > F > B > Mg > Rb

C) Rb > Mg > F > N > B

D) Mg > Rb > B > N > F

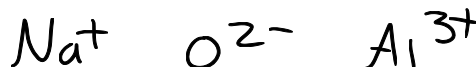
E) Rb > Mg > N > B > F

5. Two reactions are shown below, circle the one that corresponds to 1st ionization energy, or write **neither** if neither of them corresponds. (1 pt)



Ionization Energy

6. Write the ions with the correct ionic charges that are observed for monatomic ions of Na, Al, and O. (2 pts)



7. = Aktiv Chemistry Ch 3 # 135. Which of the following series of isoelectronic ions (Mg^{2+} , N^{3-} , F^- , Si^{4+}) has the ionic radii in order of largest to smallest? (1 pt)

A) $Mg^{2+} > N^{3-} > F^- > Si^{4+}$

B) $Mg^{2+} > Si^{4+} > F^- > N^{3-}$

C) $N^{3-} > F^- > Si^{4+} > Mg^{2+}$

D) $N^{3-} > F^- > Mg^{2+} > Si^{4+}$

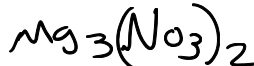
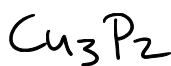
E) $F^- > N^{3-} > Si^{4+} > Mg^{2+}$

8. ~ Aktiv Chemistry Ch 3 # 142, 149. Predict & write the chemical formula for the three following ionic compounds: (2 pts)

Cu^{2+} & P^{3-}

Ag^+ & Cl^-

Mg^{2+} & NO_3^-



9. = Aktiv Chemistry Ch 4 # 35. Which of the following bonds is a polar covalent bond? (1 pt)

A) Li - Cl

B) Cl - Cl

C) H - Cl

D) Li - Li

10. = Aktiv Chemistry Ch 4 # 32. Which of the following is an ionic compound? (1 pt)

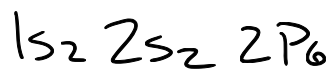
A) MgS

B) H_2S

C) S_8

D) SO_2

11. = Chapter 4 Slide 11. Write the electron configuration for the C^{4-} anion? (1 pt) $10e^{-}$



12. Name the following ionic compounds. (3 pts)

a. NaCl

Sodium Chloride

b. FeO (Fe is iron)

Iron oxide

c. $CaSO_4$ (SO_4^{2-} is 'sulfate')

Calcium Sulfate

13. We discussed this last Thursday, Bonus point! What is the ionic charge on Pb if it is in the ionic compound $PbCl_4$?

